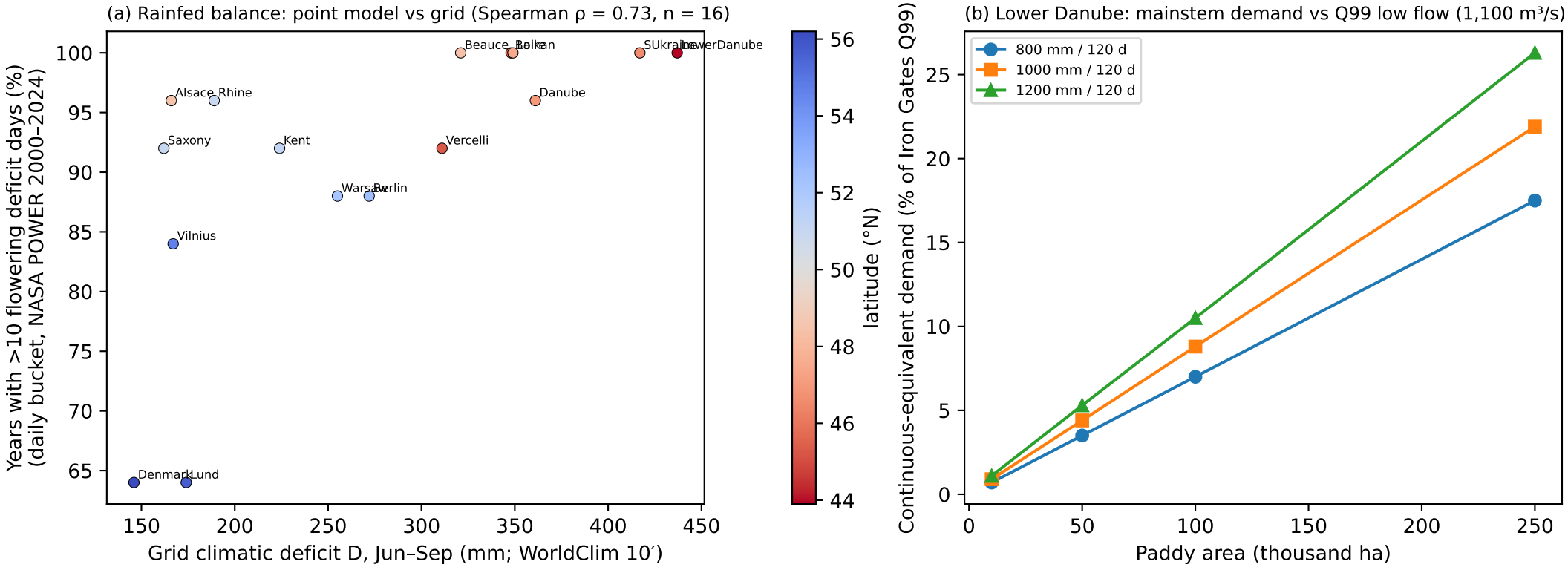
(a) Rainfed balance: point model vs grid (Spearman ρ = 0.73, n = 16)
Years with >10 flowering deficit days (%)
(daily bucket, NASA POWER 2000–2024)
Grid climatic deficit D, Jun–Sep (mm; WorldClim 10′)
Beauce
Balkan
SUkraine
LowerDanube
Alsace
Rhine
Danube
Saxony
Kent
Vercelli
Warsaw
Berlin
Vilnius
Denmark
Lund
latitude (°N)
(b) Lower Danube: mainstem demand vs Q99 low flow (1,100 m³/s)
800 mm / 120 d
1000 mm / 120 d
1200 mm / 120 d
Continuous-equivalent demand (% of Iron Gates Q99)
Paddy area (thousand ha)